

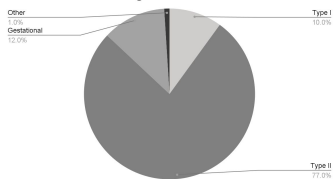
BreatheEZ: Accessory to VOC Breath Analysis Devices

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What are VOCs?

Human breath is composed of nitrogen, oxygen, carbon dioxide, and a small percentage of volatile organic compounds (VOCs), formed through lung metabolic gas exchange. Some VOCs are potential biomarkers for diseases like heart ischemia, diabetes, and lung cancer. Diabetes, in particular, exhibits elevated ketone levels in breath, particularly acetone, due to fat metabolism. Breath analytic devices offer non-invasive diabetes monitoring with reduced economic impact. While promising, VOC detection in breath poses challenges due to low concentrations (parts per billion). Current devices face limitations in specificity, selectivity, and sensitivity, underscoring the need for improved breath analytics devices to overcome existing issues.

Incidence of Diabetes Diagnoses



Direct Healthcare Cost: \$237 Billion/yr

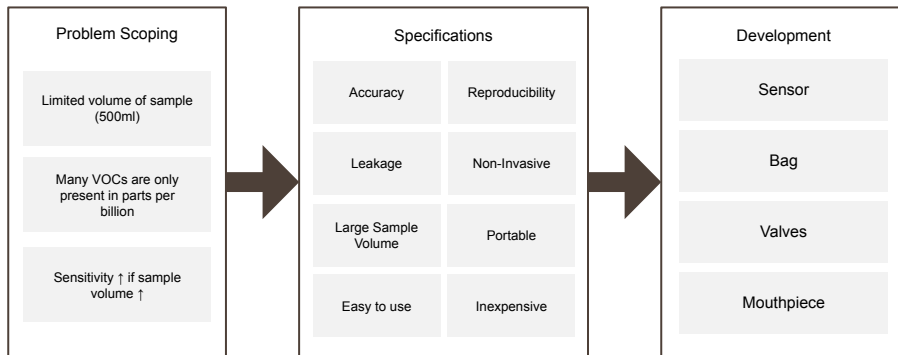
Diabetics pay about one quarter of healthcare expenditures in the United States, while only making up 11.3% of the population.

Diabetes management: \$16,752/year/person

Glucose monitoring device and strips: \$4,380/year/person

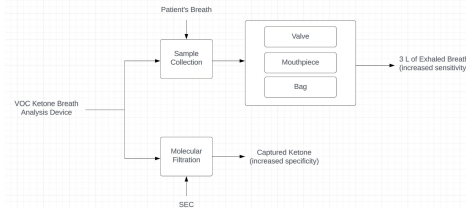
Glucose monitoring strips used daily: 4-10 strips/person

Approach

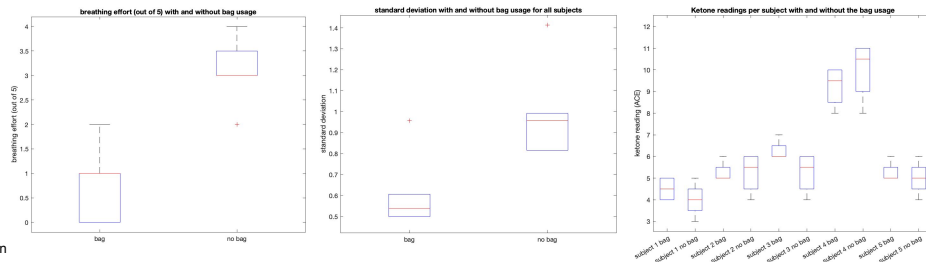


Clinical Need

There is a clear need for enhanced breath analytic devices that can effectively address sensitivity, selectivity, and accessibility issues in VOC monitoring devices.



Statistical Results



Future Applications

Our results demonstrated consistent performance across various conditions, aligning with our objective to deliver an accessible and effective solution for monitoring ketones. It is recommended to explore further refinements in design and functionality. This might encompass adjusting accessories to match specific VOC profiles linked to various diseases, continuing validation using headspace techniques or gas chromatography, and optimizing the number of sensors to minimize energy consumption while enhancing efficiency. Moreover, exploring the incorporation of advanced machine learning algorithms for real-time analysis has the potential to significantly augment the precision and efficiency of the monitoring process. Future modifications are encouraged when more VOC devices are released to market.